Mmr

Maximal Marginal Relevance for Diversity-Based Reranking
Jaime Carbonell and Jade Goldstein (SIGIR 1998)

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What the paper proposes

MMR balances query relevance against similarity to already selected passages. It is designed to reduce repeated or near-duplicate evidence while preserving useful coverage.

What changed in this project

Applied conservative MMR to the dense top-100 candidates with lambda=0.95. The implementation uses the existing chunk vectors and greedily selects a non-redundant top 20.

Measured results

Metric	Baseline	Method	Change
Hit@1	80.0%	80.0%	+0.0 pp
Hit@5	85.0%	80.0%	-5.0 pp
Recall@5	64.7%	60.9%	-3.8 pp
NDCG@10	84.0%	82.4%	-1.5 pp
MRR@20	82.9%	82.9%	+0.0 pp
Latency	26.9 ms	181.1 ms	—

Inference

MMR can improve rank diversity, but exact gold-label recall did not improve here. This is a useful negative result: the benchmark labels many duplicate manual chunks as relevant, while diversity intentionally removes some duplicates.